AIOps for traditional SAP ERP Systems

Manage the challange between traditional ITIL based operation in a where with introducing AI functionality

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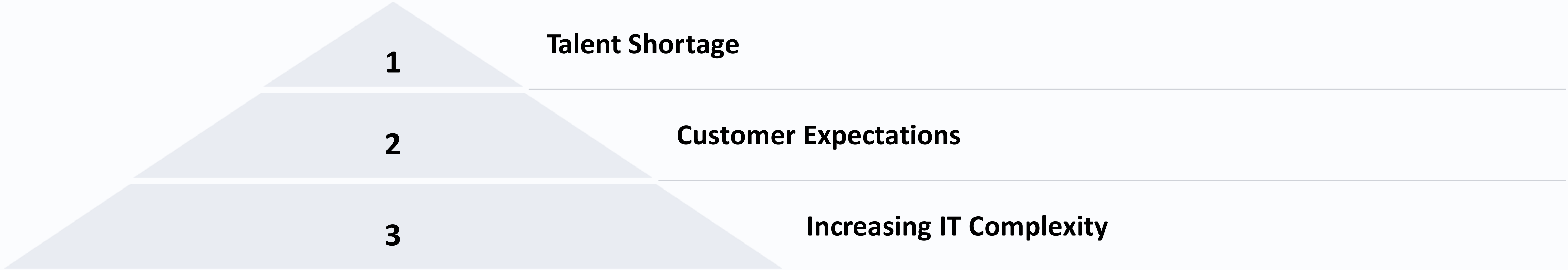
Drivers for Change: Why transform to SRE with AIOps?



The exponential growth in infrastructure complexity and scale makes traditional manual monitoring impossible. Modern enterprises manage thousands of servers, containers, and microservices across hybrid environments.

Customers now demand 99.99% availability and instant resolutions, with digital services expected to function flawlessly 24/7. Meanwhile, a critical shortage of skilled operations personnel makes it increasingly difficult to staff round-the-clock operations teams.

Traditional Operations: Pros and Cons

Advantages

Mature Processes

Well-established ITIL frameworks and documented procedures provide a comfortable operational foundation.

Predictable Operations

Lower short-term risk due to familiarity with existing systems and established troubleshooting methods.

Known Governance

Clear approval chains and documentation trails satisfy compliance requirements.

Limitations

Poor Scalability

Manual processes break down as infrastructure grows, requiring proportional staffing increases.

High Operational Costs

Labour-intensive operations drive up costs while providing limited business value.

Innovation Barriers

Teams spend 70%+ time on maintenance rather than innovation, hindering competitive advantage.

Enter AIOps: Accelerating Transformation

AIOps (Artificial Intelligence for IT Operations) represents the next evolution in enterprise IT management, combining big data, analytics, and machine learning to automate and enhance IT operations processes.

Data Collection

Ingests metrics, logs, and events from across the IT ecosystem

Automation

Executes remediation workflows without human intervention

Analysis

Applies ML algorithms for pattern recognition and anomaly detection

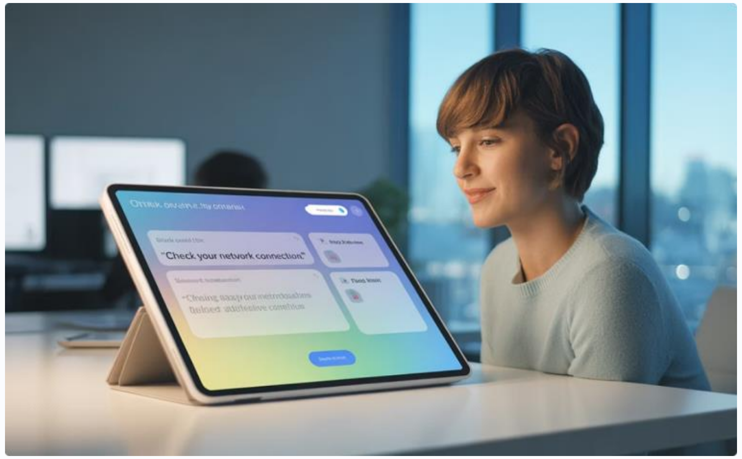
Correlation

Identifies relationships between seemingly disparate events

Root Cause

Determines probable causes and suggests remediation actions

AIOps Capabilities: LLMs, Automation & Integration

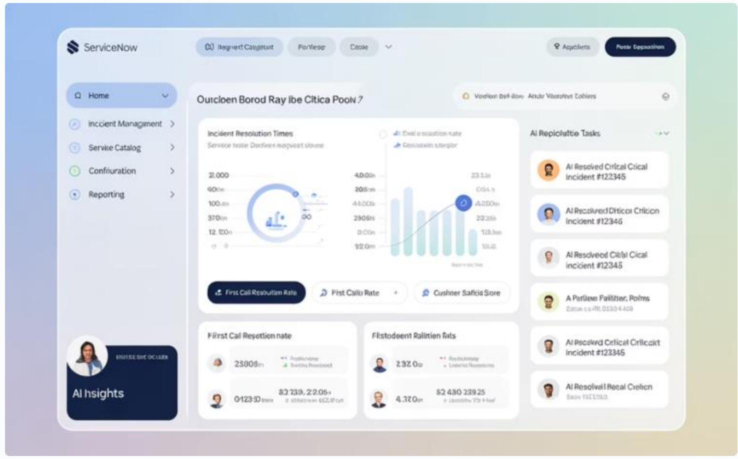


LLM-Powered Assistance

Large Language Models provide context-aware support for complex troubleshooting, generate documentation, and enable natural language interfaces for operations teams.

Runbook Automation

Intelligent playbooks automatically execute complex remediation procedures, reducing human error and accelerating resolution timeframes from hours to minutes.



ServiceNow Integration

Seamless connection with SNOW workflows enables end-to-end automation across incident management, change control, and CMDB processes while maintaining governance.

The Road to AIOps

Platform

- Transform collection and aggregation layer globally to open standards, considering workload specifics and prevent vendor lock in wherever possible.
- One Scalable AIOps infrastructure covering all data privacy requirements in a secure way
- Continuous increase coverage of metrics, logs, events data volume stored as clean data with standardization

Traditional Operation

- Isolated teams collaborating
- Limited data analysis capability without E2E view
- Reactive maintenance
- Automation islands, no holistic process integration

Processes

- Standardization and operational data integrated into processes ensure consistent and clean data as successfactor for maschine learnig trainings
- Standarized service delivery using a sufficent CI structure covering the scope of services delivered
- Adapted ITSM process simplified for less human interaction

People

- Skilled People with broad technical knowledge and understanding about the services provided toward the customer in a SRE model
- Continuous monitoring of results and adjustments if required according clear goals and KPIs
- Effective Change Management during AIOps introduction to overcome obstacles and get acceptance in the teams adapting the SRE model

AIOps Service Delivery

Integrated teams collaborating base on one single source of truth
Fact based decission on analysed data

- Maschine Learning and AI continiously adapt and improve the pattern analyss
- Proactive issue detection to prevent Incidents, ensure business continiouty wit proactive maintenance

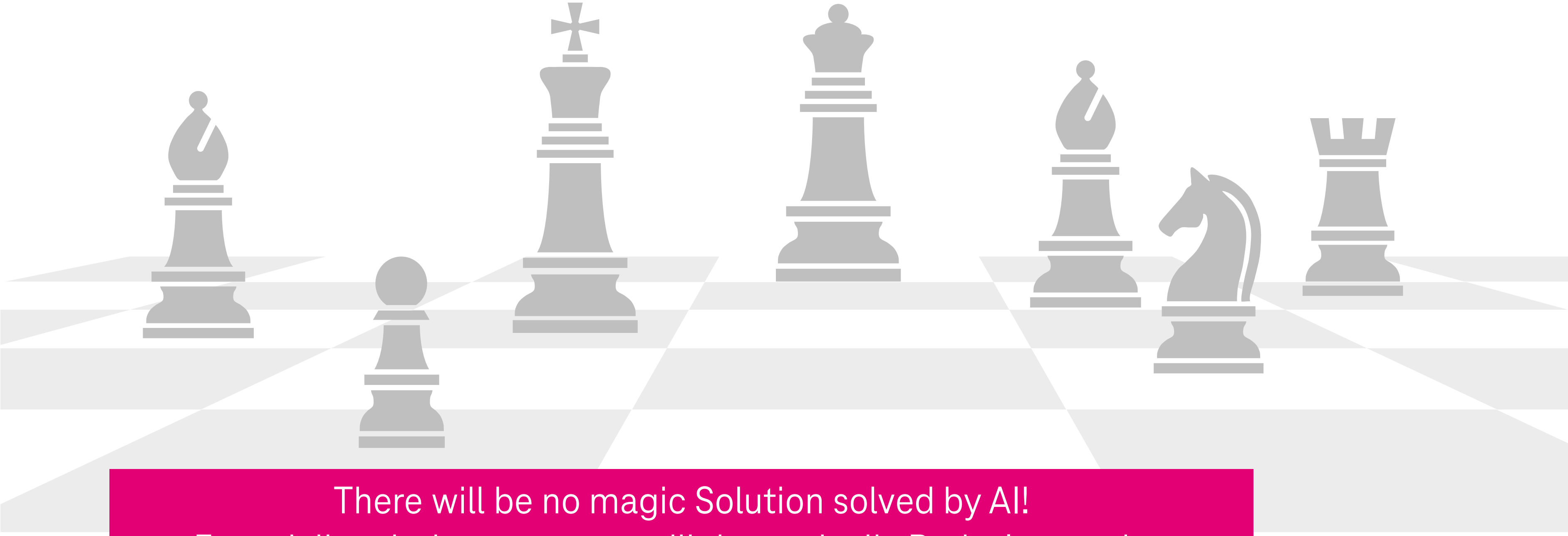
Tactical Challenges for Managed Service Provider

Foster Multi and Hybrid Cloud Deployment Standards

- Extend Standards to exist reach feature completeness and support service delivery automation
- Improve ITSM integration for real E2E services starting with the basics like technical Cis, clear Service chains
- Distinguish classic Managed Service delivery and new services provided as Cloud Operation model
- Ensure New Services as Default offering
- Non-Standard business as customer tailored manufacturing solution with individual delivery approval and costing

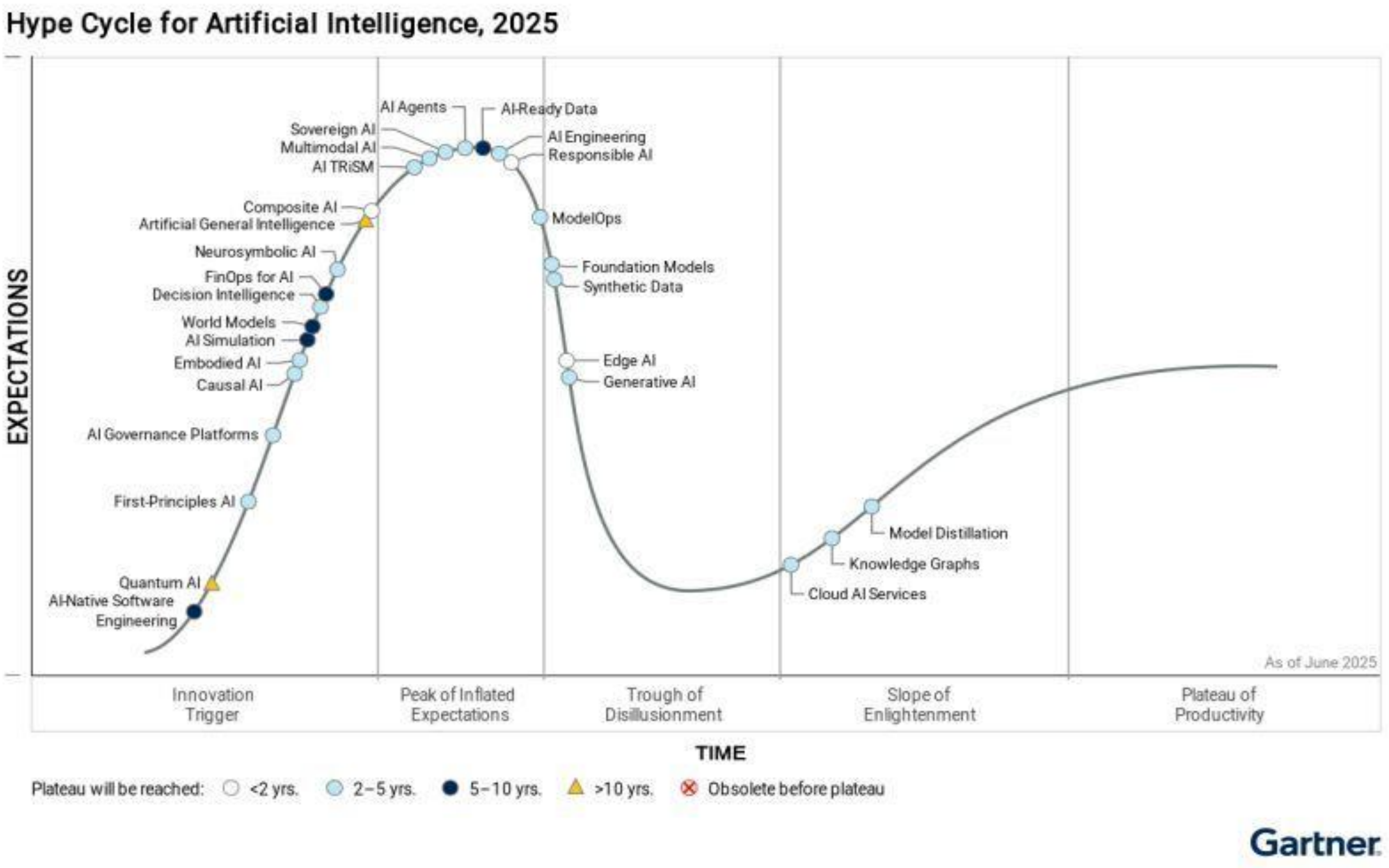
People and Process Transformation

- Several Organizational changes Overlap during people transformation
- Resistance to change in Tools, Organization and service scope
- Lost knowledge in operational basics
- Increased complexity of services and dependency’s
- The amount of technology frameworks used to provide services continuously growing



There will be no magic Solution solved by AI!
Especially missing structure will dramatically Brake Innovation

What is about AI



Agentic AI (Systems that perform tasks without human intervention)

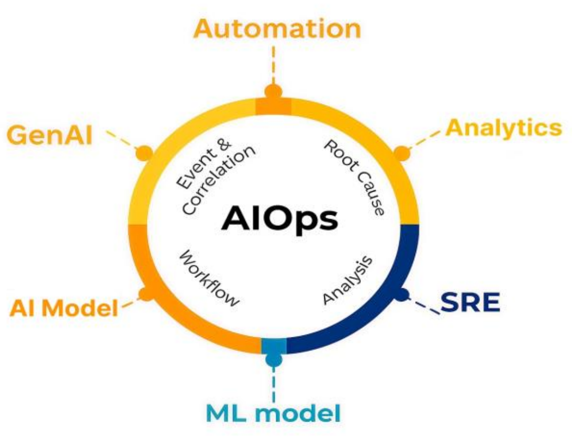
Agentic AI is a class of artificial intelligence that focuses on autonomous systems that can make decisions and perform tasks without human intervention. The independent systems automatically respond to conditions, to produce process results. (AIOps)

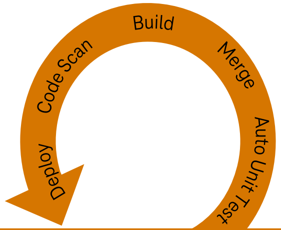
Responsable AI, TRiSM (Trust, Risk & Security Mgmt) and AI-ready Data become focus
The path to truly productive AI is not through hype, but through architecture, pragmatism and clean operational data.

Build & Run Collaboration Model

Aligning Agile Development with ITSM Standards to Drive Agility, Operational Discipline and Excellence

BUILD

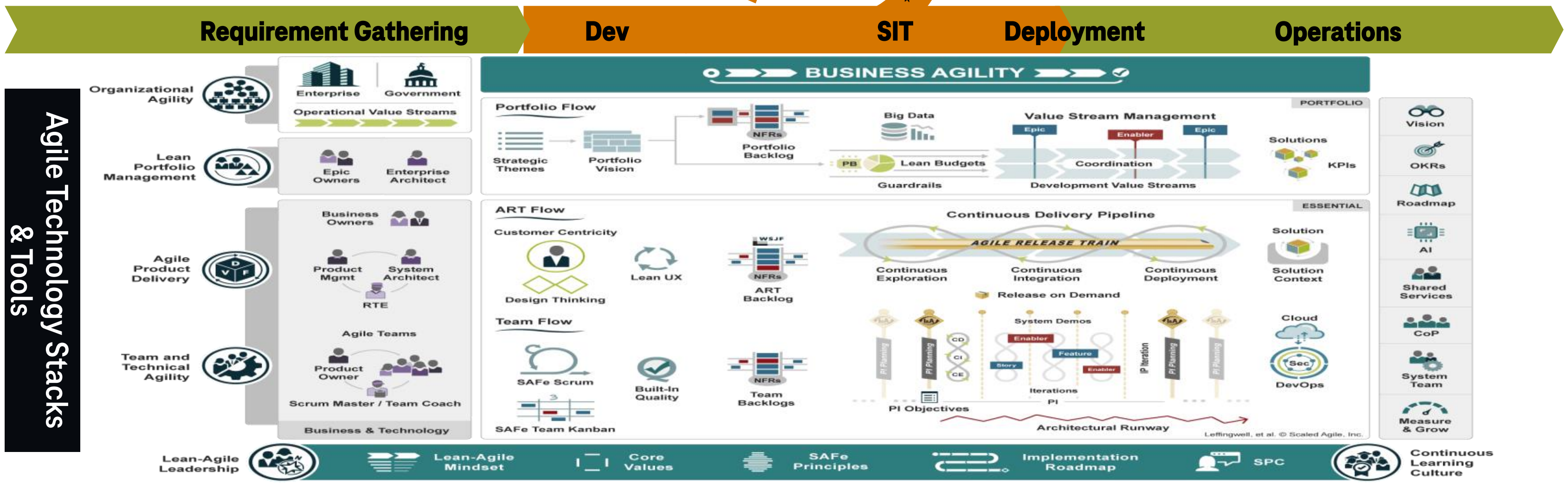




RUN

servicenow

ITSM2.0



ITSM still the foundation

- To fulfill Audit and Compliance Regulations
- Regulatory Prerequisite for Kritis and Stock Market listed Enterprises

Classical Dev and Operations – The Problem

Classical Dev and Operations - Disadvantages

Who is feeling the pain?

Developers
- Frequently disagree with system administrators
- Spending time on mundane manual tasks

Test Engineers
- Frequently disagree with system administrators
- Receive differing instructions from developers" code and operations' directions

Operations
- Challenging audits
- Hard-to-diagnose outages
- Receive code from developers with little communication

QA
- Dealing with slow error-prone software delivery
- Spending copious amounts of time fixing costly errors

Product Managers
- Frequently disagree with system administrators
- Costs and inefficient projects

Customer Service
- Trying to placate disgruntled customers
- Stuck in the middle of a system with little communication

System Administrators
- Frequently disagree with developers and others
- Lack visibility in build, deploy, test, and release processes
- Considerable time spent on upkeep

End Users
- Frequent and prolonged outages
- Hard-to-navigate systems

Tailored AIOps for Enterprise Customers

Our distinct expertise lies in transforming traditional operations to a GitOps-driven SRE model, specifically addressing the complex and often highly customised environments of enterprise customers.

Challenge: Enterprise Customisation

Standard automation falls short for highly unique customer setups, leading to manual workarounds and inefficiency.

AI-Powered Custom Automation

Operations teams leverage AI Code Pilots (e.g., for Ansible, Terraform) to rapidly develop bespoke automation for these 'specials'.

GitOps-Driven SRE

All custom automation is integrated and managed via a GitLab-driven GitOps approach, ensuring version control and consistency.

Seamless Managed Services

This enables us to seamlessly onboard and manage even the most customised customer environments as a consistent managed service.

Future State: NoOps v2 with AIOps

NoOps represents the aspirational end-state where routine operations become invisible through complete automation, allowing IT teams to focus entirely on innovation and business value rather than maintenance.

95%

Automation Rate

Nearly all routine operations tasks handled without human intervention

99.999%

Service Availability

Five-nines reliability through predictive healing and redundancy

<1 min

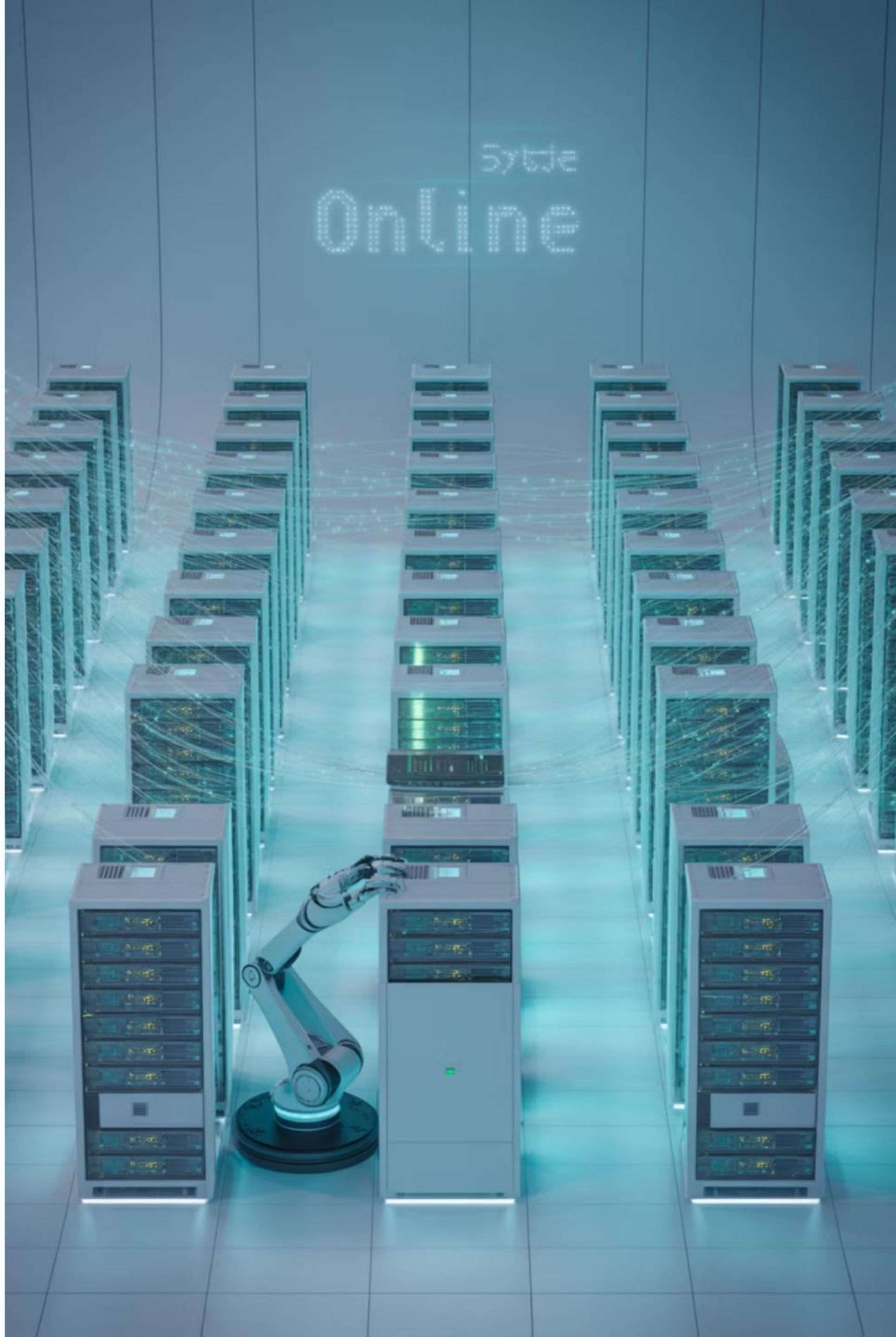
MTTR

Mean time to repair measured in seconds through autonomous remediation

80%

Innovation Time

Majority of IT staff time dedicated to new capabilities and business growth



What is SRE?

Site Reliability Engineering (SRE) is a discipline that **applies software engineering principles to infrastructure and operations**, ensuring that systems are scalable, reliable, and efficient. SRE teams focus on automating routine tasks, based on reusable standards and technology available. Key is proactive E2E IT landscape observevability detecting potential issues before customer visible incidents occur to maintain to maintain high system uptime.

Partner with the customer to support the business needs and Transformations needes due to changed business models. Their approach balances rapid development with operational stability by integrating engineering practices directly into service management.

What is NoOps?

NoOps represents a vision where the operations layer is so automated that developers can focus entirely on code and functionality. It leverages modern cloud services and automation tools to reduce or eliminate manual operational tasks. However, while it’s an appealing concept, many organizations find that a hybrid approach—combining the benefits of automation with some level of human oversight—is often the most practical solution in real-world scenarios.

In essence, NoOps is about creating an environment where operations become a background service, enabling faster development cycles and reducing the friction between development and operational management.

Key Points about NoOps

Automation-Driven Infrastructure:

- In a NoOps environment, the infrastructure is designed to be self-managing through extensive automation.
- Routine tasks such as provisioning, configuration, deployment, monitoring, scaling, and even incident remediation are handled by automated tools and platforms.

Developer Empowerment:

- The goal is to allow developers to focus solely on writing and deploying code without having to worry about the underlying operational complexities.
- Developers can deploy applications directly, often through self-service interfaces provided by cloud or platform services.

Contrast with DevOps

- **DevOps** is a cultural and technical movement that promotes collaboration between development and operations teams, often integrating continuous integration and continuous delivery (CI/CD) pipelines.
- **NoOps** takes this further by automating the operations side so completely that a separate operations team becomes unnecessary.
- In practice, while DevOps emphasizes collaboration between two traditionally separate groups, NoOps aims to remove the boundary by automating operations tasks.

What is NoOps v2?

- By integrating AI into NoOps, you can create a more resilient, adaptive, and intelligent operational environment. AI-driven NoOps v2 can detect and resolve issues before they impact end-users, optimize resource allocation in real time, and continuously learn from the environment to improve its decision-making processes. This not only streamlines operations but also bridges gaps in expertise, enabling organizations to maintain high performance and reliability even as complexity increases.

- Ultimately, NoOps v2 represents a paradigm shift where operations become a dynamic, self-optimizing ecosystem—allowing teams to focus on innovation while AI handles the heavy lifting of day-to-day management.

Intelligent Anomaly Detection and Monitoring

Real-Time Insights:

- AI-powered monitoring systems can continuously analyze logs, metrics, and events in real time, detecting anomalies and performance deviations that traditional monitoring tools might miss.

Context-Aware Alerts:

- Machine learning models can learn normal operational patterns over time and trigger alerts only when true anomalies occur, reducing noise and false positives.

Predictive Analytics:

- By analyzing historical data, AI can predict potential failures or performance bottlenecks before they occur, enabling proactive interventions rather than reactive fixes.

Automated Incident Response and Self-Healing Systems

Intelligent Root Cause Analysis:

- AI agents can quickly correlate disparate data sources (logs, metrics, user reports) to identify the root cause of an incident. This speeds up resolution and minimizes downtime.

Self-Healing Capabilities:

- When an issue is detected, AI can automatically initiate remediation procedures such as rolling back a deployment, reallocating resources, or restarting services—effectively “healing” the system without human intervention.

Reinforcement Learning for Continuous Improvement:

- By leveraging reinforcement learning, AI systems can learn from past incidents and improve their remediation strategies over time, becoming more effective at preventing and resolving issues.

Dynamic Resource Management and Optimization

Adaptive Scaling:

- AI can predict usage patterns and dynamically adjust resources (e.g., CPU, memory, storage) based on real-time demand. This not only ensures optimal performance but also minimizes costs by preventing over-provisioning.

Capacity Planning:

- With continuous analysis of resource utilization trends, AI can forecast future requirements and help plan capacity expansions or optimizations well in advance.

Load Balancing and Traffic Optimization:

- AI algorithms can intelligently distribute network traffic and workloads across servers or containers, optimizing performance and reducing latency during peak loads.

Enhanced Security and Compliance Automation

Proactive Threat Detection:

- AI can continuously scan for unusual patterns that may indicate security breaches or vulnerabilities, automating the identification and sometimes even the mitigation of security threats.

Automated Compliance Monitoring:

- AI-driven tools can ensure that the infrastructure complies with regulatory standards by continuously monitoring configurations and automatically remediating any deviations.

Incident Forensics:

- In the event of a security incident, AI can rapidly analyze data to trace the origin of the breach and suggest countermeasures, accelerating the incident response process.

Streamlined DevOps Integration and Workflow Optimization

- **Automated Deployment Pipelines:**
- AI agents can optimize CI/CD pipelines by determining the best times for deployments based on historical success rates, current system load, and predictive analytics.
- **Intelligent Rollbacks and Blue-Green Deployments:**
- In case a deployment goes awry, AI can quickly decide whether to roll back or switch to a backup environment, reducing downtime and service disruption.
- **Feedback Loop Enhancement:**
- By analyzing deployment outcomes, performance metrics, and user feedback, AI can continuously refine and optimize the automation workflows, ensuring that the system evolves to meet emerging challenges.